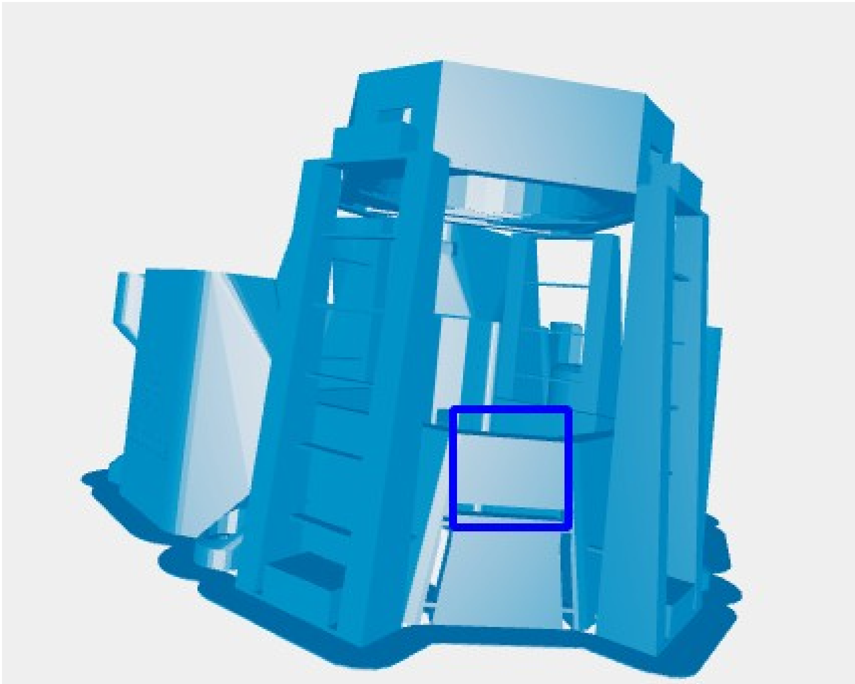
Building Your RayV Light Laser Fault Injection – Modified OpenFlexture Instructions

This guide provides modified instructions for building the OpenFlexure Microscope v7.0.0-beta2, incorporating necessary deviations from the official build guide to convert it into the Laser fault injection system. Refer to the official guide for the primary instructions:
https://build.openflexure.org/openflexure-microscope/v7.0.0-beta2/high_res_microscope.html

Deviations:

1. Main Body Assembly:

During the main body assembly, you'll need to modify the 3D printed part. The official instructions don't account for a necessary change. Specifically, **remove the brace highlighted in the blue square in the image below**.



This brace interferes with laser module as currently designed. While ideally, this model would be edited, there was some difficulty attempting to remove the brace in FreeCAD, if anyone is able to remove/move the brace and can provide a model we will test and upload it here if successful. Currently, physical removal is the recommended workaround.

2. Optics Module Assembly:

The optics module assembly also requires modifications. Refer to the official optics module instructions: https://build.openflexure.org/openflexure-microscope/v7.0.0-beta2/high_res_optics_module.html

Changes:

- Beam Splitter Housing: Instead of the standard part, use the file Openflexure_Additions/optics_picamera2_rms_f50d13_beamsplitter_delta.stl. This modified housing is designed to accommodate the specific beam splitter you'll be using.
- Beam Splitter Cube: Use the file Openflexure_Additions/fl_cube.stl for the beam splitter cube. The final assembled cube should resemble the image below.

For detailed instructions please follow the instructions on this page steps 6-11. https://build.openflexure.org/openflexure-delta-stage/v1.2.0/pages/reflection_illumination.html

Instead of step 12, use the laser tube found here: Openflexure_Additions/laser_holder.stl.

It should attach similarly to the the excitation filter does in step 12.

- Beam Splitter Installation: Carefully insert your purchased beam splitter (see the spreadsheet for description and link) *inside* the 45-degree clip of the fl_cube.stl part. **Important: You may need to carefully cut the glass of the beam splitter to ensure it fits correctly within the clip**.
- Optics Tube and Laser Holder**:** After installing the beam splitter in the cube, insert the assembly into the optics tube. Attach the laser holder using the file Openflexure_Additions/laser_holder.stl. The pieces should just slide together. If they do not, we recommend using black Sugru to block out any remaining light. The shaft of the laser holder should fit snugly into the slot you created by removing the brace from the main body (step 1).

3. Remaining Assembly:

Once you have completed the above modifications, continue following the remaining instructions in the official OpenFlexure build guide.